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Research Report

AI - Model Architecture Selection

v1.0

Comparative Analysis and Recommendation of Deep Learning Architectures for Automatic Glaucoma Detection in Fundus Imaging

Project Subject: Glaucoma detection using intelligent analysis of fundus images

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Contents

Executive Summary

This document is a comprehensive response to the project's R&D need for automatic glaucoma classification (healthy/diseased) from fundus images. Its goal is to firmly resolve the dilemma of the technological paradigm - classical machine learning versus neural networks - and to select three optimal model architectures that, in light of the 2023-2025 scientific literature, offer the highest diagnostic potential.

Based on an in-depth analysis of over a hundred scientific sources, the report confirms the indisputable supremacy of Deep Learning (DL) over traditional methods, pointing to its ability to extract hierarchical features and its robustness to the variability of clinical data.

Three models are recommended, representing different architectural philosophies and addressing different deployment challenges:

1. **Swin Transformer (Shifted Window Transformer)** - a representative of the latest generation of transformer networks, offering unmatched global-context analysis.
2. **EfficientNetV2** - a peak achievement among convolutional neural networks (CNNs), providing an optimal balance between speed and precision, ideal for resource-constrained environments.
3. **GONet (based on DINOv2)** - a novel approach using Foundation Models and self-supervised learning, solving the critical problem of generalisation across diverse ethnic populations.

1 Introduction: Clinical Context and Technological Justification

1.1 Glaucoma as a Diagnostic Challenge and the Role of AI

Glaucoma, the second most common cause of irreversible blindness worldwide, is an enormous challenge for healthcare systems; epidemiological estimates indicate that by 2040 the number of affected people will exceed 111 million. The disease is a progressive optic neuropathy leading to apoptosis of retinal ganglion cells and irreversible visual-field loss.

Traditional clinical diagnosis relies largely on assessing optic nerve head (ONH) morphology and the CDR (Cup-to-Disc Ratio). However, this process is highly subjective: studies show that even experienced specialists may disagree on the same case, and early-stage (preperimetric) glaucoma may not manifest as obvious CDR changes but as subtle changes in the retinal nerve fibre layer (RNFL) or vascular asymmetry.

In this context, AI systems are not meant merely to replicate the physician's decision, but to *superimpose* visual information, enabling the detection of patterns imperceptible to the human eye. Automating this process is crucial for large-scale screening where access to specialists is limited.

1.2 Resolving the Dilemma: Are Neural Networks the Best Choice?

In light of current knowledge, the answer is unequivocally yes. To understand why, one must look at the evolution of image-processing methods in ophthalmology.

1.2.1 Limitations of Classical Methods (Feature Engineering)

Before the Deep Learning era, handcrafted-feature methods dominated, using image-processing algorithms to extract human-defined parameters such as:

- Image texture (analysed with LBP - Local Binary Patterns, or GLCM - Gray Level Co-occurrence Matrix).
- Colour and pixel-intensity histograms.
- Blood-vessel morphology.

These features were then fed to classical classifiers such as SVM, Random Forest, or AdaBoost. Although in highly controlled conditions such methods achieved high scores (e.g. LBP + GLCM with SVM reached 98.58% accuracy in one study), their main weakness is a lack of robustness to data variability. They “see” only what the engineer told them to measure. When an image is noisy, under-exposed, or the pathology is atypical (e.g. no clear CDR enlargement but disc haemorrhages present), classical algorithms fail - they cannot generalise or “understand” the image as a whole.

1.2.2 The Supremacy of Deep Neural Networks

Neural networks, in particular Convolutional Neural Networks (CNNs) and the latest Vision Transformers (ViT), changed the paradigm. Instead of relying on predefined features, they learn data representations directly from raw pixels, hierarchically:

- **Shallow layers:** learn simple geometric structures - edges, lines, colour gradients.
- **Intermediate layers:** compose these into more complex structures - vessel fragments, optic disc arcs, areas of atrophy.
- **Deep layers:** integrate this into high-level semantic concepts, distinguishing a “healthy” from a “glaucomatous” eye based on non-linear correlations of thousands of microscopic features.

Comparative analyses from 2024-2025 confirm that Deep Learning models achieve higher AUC, sensitivity, and specificity than any classical method. Moreover, unlike classical methods, neural network performance grows logarithmically with the amount of training data (the scale effect), making them the only valid choice for a modern medical project. For example, a VGG19 model reached 99.43% accuracy in a direct comparison, outclassing older approaches.

Conclusion: abandoning neural networks for classical methods in 2025 would be a substantive and technological mistake, condemning the project to lower performance and no room for growth.

2 Technology Landscape 2024-2025: State-of-the-Art Review

Between 2015 and 2020, architectures such as ResNet50, InceptionV3, and VGG16 were the standard. They remain solid and are still often used as benchmarks, but in the most recent studies they yield to newer solutions in efficiency and accuracy. In 2025, three main trends lead the field:

1. **Vision Transformers (ViT):** self-attention models that capture long-range dependencies in an image, surpassing CNNs in tasks requiring global context.
2. **Efficient CNNs (EfficientNets):** an evolution of convolutions toward maximum efficiency with minimal parameters, allowing training at higher resolutions.
3. **Foundation Models and SSL (Self-Supervised Learning):** moving away from training from scratch on small medical sets toward powerful models pretrained on huge unlabelled image collections (e.g. DINOv2), which drastically improves generalisation.

The three recommended models below were chosen to represent the best solution from each trend.

3 Recommended Model #1: Swin Transformer (Shifted Window Transformer)

3.1 Architecture and Innovation

The Swin Transformer is a breakthrough in adapting the Transformer architecture to vision tasks. Whereas the classical ViT required enormous compute (quadratic complexity in the number of pixels), Swin introduced the **Shifted Windows** mechanism, working in two stages:

1. The image is divided into windows, and self-attention is computed only locally within them, reducing complexity to linear - comparable in efficiency to a CNN.
2. In subsequent layers the window grid is shifted, enabling information exchange between neighbouring windows. The model thus builds a hierarchical image representation while retaining the ability to model global relations.

This is crucial for fundus images: glaucoma is not a point disease. Local changes in the optic disc must be analysed in the context of the vascular-bundle thickness elsewhere in the retina (global). The Swin Transformer links such distant pixels far more effectively than classical CNNs with their limited receptive field.

3.2 Results and Evidence

- A study combining Swin with the SegFormer architecture achieved 97.8% accuracy, surpassing VGG-19 and DenseNet169.
- In a direct accuracy-vs-FLOPs comparison, Swin-B (Base) reaches 86.4% top-1 accuracy on ImageNet - 2.4% better than the classical ViT, at lower computational cost.

- In medical segmentation and classification, Swin shows higher precision in detecting subtle textural changes, critical for early glaucoma diagnosis.

3.3 Deployment Variant: Swin-B (Base)

The Swin-B variant is recommended: $\sim$88 million parameters and 15.4 GFLOPs. It is the "sweet spot" - large enough to learn complex glaucoma patterns, yet trainable on a single powerful GPU. Smaller variants (Swin-T, Tiny) may be considered for very limited resources, but Swin-B offers the best quality-to-cost ratio.

4 Recommended Model #2: EfficientNetV2 (SOTA Convolutional Network)

4.1 Architecture and Innovation

If the Swin Transformer is the king of context, EfficientNetV2 is the master of engineering and efficiency. It is a family of convolutional models born from Neural Architecture Search, optimised for training speed and precision. Key V2 innovations over V1 include:

- **Fused-MBConv:** replacing slow depthwise convolutions in early layers with fused convolutions, drastically speeding up processing on modern GPU/TPU accelerators.
- **Progressive Learning:** dynamically changing image size and regularisation during training - small images and weak augmentation early (fast), large images and strong augmentation late (precise).
- **Compound Scaling:** intelligent, simultaneous scaling of depth, width, and resolution, preventing information-flow bottlenecks.

4.2 Results and Evidence

- Practitioners report that EfficientNetV2-L achieved an F1 of 91.3% in retinopathy detection under difficult clinical conditions (poor lighting, cheap cameras), a 42% inference-latency reduction versus ResNet152 - evidence it excels in real-world, not just sterile lab, data.
- On OCT datasets (a related imaging technique), EfficientNetV2 reached 98.8% accuracy in classifying healthy cases, outclassing older ViT models.
- On the ACRIMA dataset, EfficientNet-family models routinely exceed 97.5%, surpassing older Inception or ResNet architectures.

4.3 Deployment Variant: EfficientNetV2-S (Small)

EfficientNetV2-S is recommended to start: "only" 21.4 million parameters, making it light and very fast to train. In medicine, where datasets rarely exceed a few tens of thousands of images, giant models (V2-L, V2-XL) can overfit. The "Small" variant offers strong expressive power while preserving generalisation safety.

5 Recommended Model #3: GONet (DINOv2 / Foundation Model)

5.1 Architecture and Innovation

GONet is the most novel proposal, based on 2025 publications. It is a glaucoma-dedicated model (Glaucoma Optimized Network) that uses the powerful DINOv2 as its backbone. DINOv2 is a Vision Transformer trained under the Self-Supervised Learning (SSL) paradigm: it "saw" millions of unlabelled images and had to learn the structure of the visual world (distinguishing objects from background, understanding depth and texture) to solve pretext tasks. Unlike ImageNet-supervised models, DINOv2 produces far more universal and robust features.

5.2 Why It Is a "Game Changer" for Glaucoma

The main problem of medical AI is **Domain Shift**: a model trained on Asian fundus photographs (e.g. REFUGE) often performs poorly on European ones (e.g. RIM-ONE) due to differences in fundus pigmentation. GONet solves this via a **Multi-Source Domain Adaptation** strategy - it was trained on seven independent datasets. Thanks to this and the powerful DINOv2 backbone, it achieves unprecedented generalisation.

5.3 Results and Evidence

- GONet showed an AUC of 0.85-0.99 on target domains it was *not* trained on - an outstanding result, proving the model learned "what glaucoma is", not "what the training set looks like".
- In comparisons, GONet beat a CDR-based method by 21.6%, finally disqualifying simple morphological methods.
- Code and weights availability (authors: Abramovich et al., Technion) allows using this model as a powerful starting point (transfer learning) for our own project.

6 Comparative Analysis and Rejected Alternatives

To make the choice complete, here is why other popular models did not make the top three.

6.1 MobileNetV2 vs. DenseNet121 vs. ResNet50

- **MobileNetV2:** excellent for mobile use, reaching ∼81.3%-84.5% accuracy. Decent, but clearly below the leaders. Unless the project requires running on a smartphone without Internet, MobileNetV2 offers too little capacity to detect subtle glaucoma changes.
- **DenseNet121:** excellent feature reuse, often better than MobileNetV2 (90.93% vs 84.54%) and more stable on small datasets. A strong "#4" candidate, but EfficientNetV2 usually offers an even better performance-to-memory ratio.
- **ResNet50:** a classic, with solid results (∼87-90%), but 2015 technology. Modern models match it with fewer parameters or beat it at the same cost. Use ResNet only as a baseline to validate the data pipeline.

Feature	**Swin Transformer (Swin-B)**	**EfficientNetV2 (V2-S)**	**GONet (DINOv2)**
Architecture type	Vision Transformer (hierarchical)	CNN (convolutional)	Foundation Model (SSL ViT)
Parameters	∼88 M	∼21.4 M	∼86 M (ViT-B backbone)
GFLOPS (cost)	∼15.4 G	∼8.4 G	High (input-dependent)
Main advantage	Global context: best analysis of spatial relations	Efficiency: fastest training, low memory	Generalisation: best robustness across datasets
Recommended use	Maximising results (SOTA) on a GPU server	Fast iterations, low-latency production	Multi-centre deployment, hard/diverse data

Table 1: Comparison of the recommended models.

7 Implementation and Training Strategy (Best Practices)

Choosing the model is half the battle; in medical projects, data preparation and the training procedure are decisive.

7.1 Data Preprocessing

- **Region-of-Interest (ROI) extraction:** the background (black corners, off-eye artefacts) is redundant. Automatically detect the optic disc (e.g. with a simple U-Net) and crop a square around it (e.g. 512×512). Models fed only the ROI often perform better.
- **Illumination correction (CLAHE):** Contrast Limited Adaptive Histogram Equalization is mandatory - it equalises local contrast and highlights fine vessels that fade in under-exposed images.
- **Green channel:** in RGB imaging, the green channel carries the most contrast about retinal structures. It is worth experimenting with feeding only this channel (or an enhanced version) to reduce chromatic noise.

7.2 Datasets and Validation

Reliable results cannot rely on a single dataset. Use publicly available bases:

- **ACRIMA:** a popular set (309 healthy / 396 glaucoma) where models often exceed 98%. Good to start.
- **RIM-ONE:** much harder; model performance often drops to 80-90%. An excellent overfitting test.
- **REFUGE:** a challenge dataset, also containing OCT images. A standard in scientific comparisons.

Cross-Dataset Evaluation: train on one set (e.g. ACRIMA) and test on another (e.g. RIM-ONE). Only this reveals whether the model learned to recognise glaucoma or just the specifics of the ACRIMA camera.

7.3 Loss Functions and Training

Move away from standard Cross-Entropy toward PolyLoss or Focal Loss. These let the model focus on hard cases and cope better with class imbalance, typical in medicine (more healthy than diseased).

8 Model Explainability (XAI) - Building Trust

Deploying medical AI requires not only performance but transparency: the physician must know *why* the model made a decision. The project should mandatorily include an Explainable AI (XAI) module. The most effective technique for CNNs is **Grad-CAM**, generating heatmaps overlaid on the eye image. The heatmap should point at the optic disc and peripapillary atrophy areas. If the model "looks" at off-eye artefacts, Grad-CAM reveals it, allowing quick correction of training errors. For the Swin Transformer and GONet, **Attention Maps** can be visualised, offering even higher interpretive resolution.

9 Summary and Action Plan

1. **Technology confirmation:** neural networks are unquestionably the best tool for glaucoma detection. Classical methods lack the flexibility and precision needed for the complexity of glaucoma pathology.
2. **Recommendation:** the team should focus on testing three models - Swin Transformer (Base) to maximise results, EfficientNetV2 (Small) to optimise speed, and GONet (DINOv2) to ensure generalisation.
3. **Key to success:** architecture is not everything. Preprocessing (CLAHE, ROI cropping) and a multi-dataset training strategy (cross-dataset validation) are critical to avoid a solution that works only in laboratory conditions.

Following these recommendations will enable a world-class diagnostic system aligned with the latest 2024-2025 scientific standards.

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